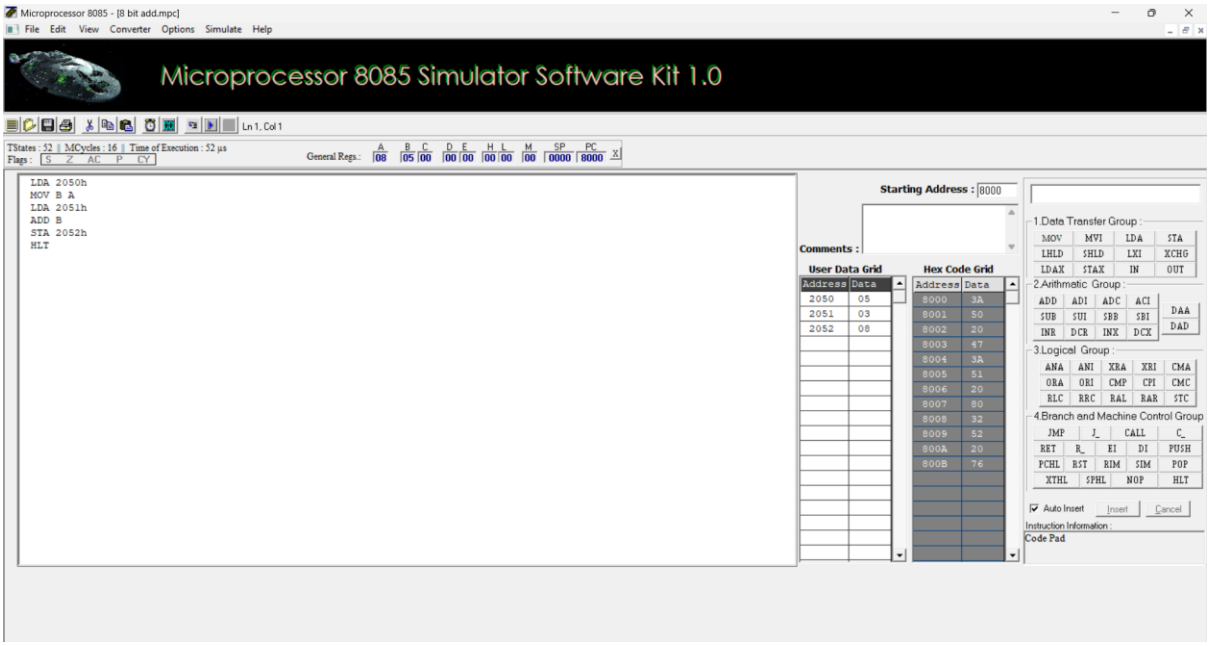
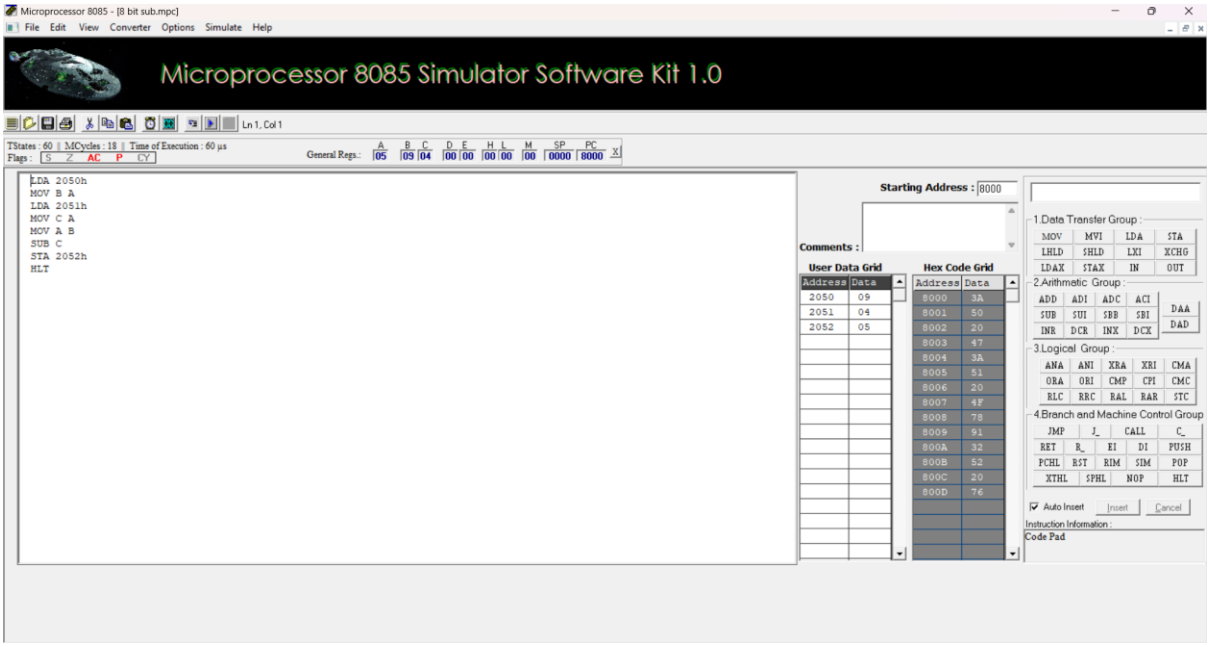
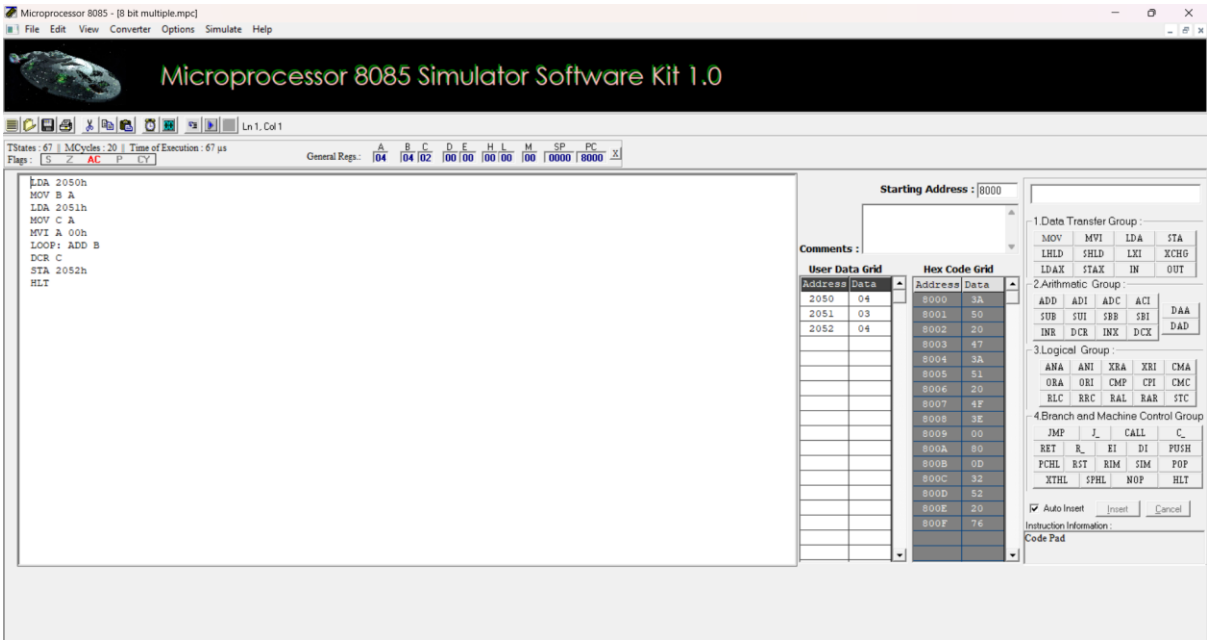
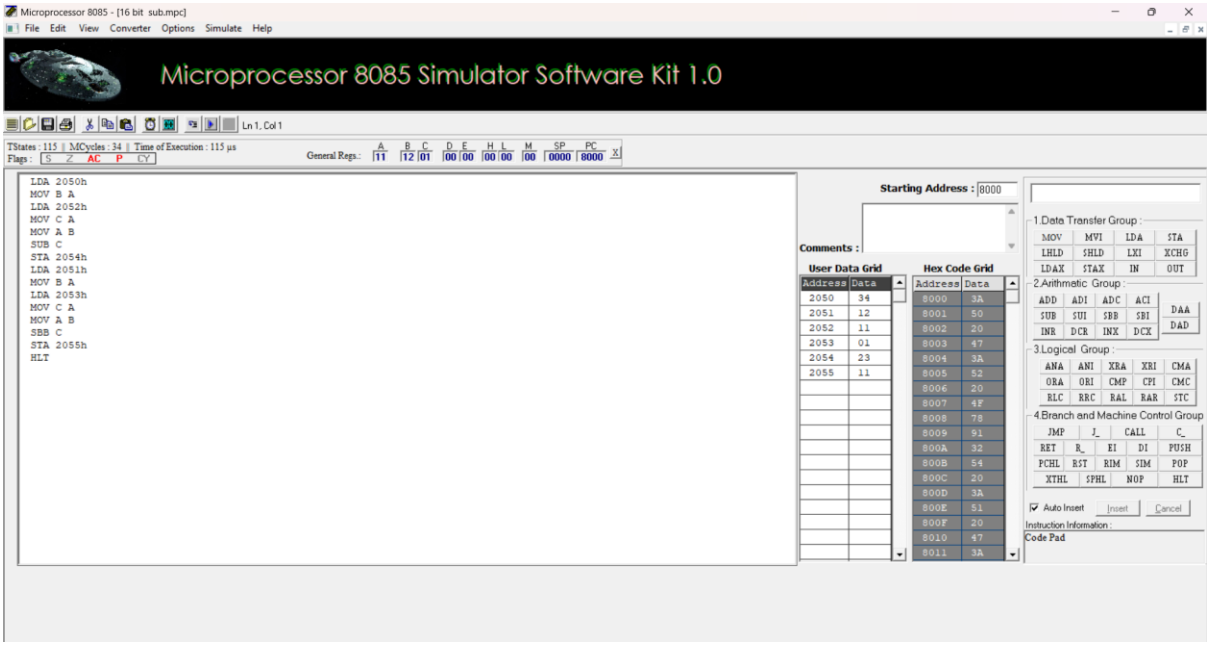
ECA 10-MICROPROCESSORS AND MICROCONTROLLERS

1. Analysis, Development and Verification of 8-bit and 16-bit Arithmetic Operations Using Assembly Language Program Major Topics / Minor Topics Included: • 8-bit addition and subtraction• 16-bit addition and subtraction• 8-bit multiplication and division• 16bit multiplication and division• Carry, borrow and overflow• Flag-register analysis• Immediate, register and memory addressing






Microprocessor 8085 - [8 bit multiple.mpc]
File Edit View Converter Options Simulate Help
Microprocessor 8085 Simulator Software Kit 1.0
Ln 1, Col 1
TStates : 67 | MCycles : 20 | Time of Execution : 67 µs
Flags : S Z AC P CY
General Regs.: A 04 B 04 C 02 D 00 E 00 H 00 L 00 M 00 SP 0000 PC 8000
LDA 2050h
MOV B A
LDA 2051h
MOV C A
MVI A 00h
LOOP: ADD B
DCR C
STA 2052h
HLT
Starting Address : 8000
Comments :
User Data Grid
Address Data
2050 04
2051 03
2052 04
Hex Code Grid
Address Data
8000 3A
8001 50
8002 20
8003 47
8004 3A
8005 51
8006 20
8007 4F
8008 3E
8009 00
800A 80
800B 0D
800C 32
800D 52
800E 20
800F 76
1.Data Transfer Group :
MOV MVI LDA STA
LHLD SHLD LXI XCHG
LDAX STAX IN OUT
2.Arithmetic Group :
ADD ADI ADC ACI
SUB SUI SBB SBI
INR DCR INX DCX
DAA
DAD
3.Logical Group :
ANA ANI XRA XRI CMA
ORA ORI CMP CPI CMC
RLC RRC RAL RAR STC
4.Branch and Machine Control Group
JMP J_ CALL C_
RET R_ EI DI PUSH
PCHL RST RIM SIM POP
XTHL SPHL NOP HLT
Auto Insert
Insert
Cancel
Instruction Information :
Code Pad


Microprocessor 8085 - [16 bit sub.mpc]
File Edit View Converter Options Simulate Help
Microprocessor 8085 Simulator Software Kit 1.0
Ln 1, Col 1
TStates : 115 | MCycles : 34 | Time of Execution : 115 µs
Flags : S Z AC P CY
General Regs.: A 11 B 12 C 01 D 00 E 00 H 00 L 00 M 00 SP 0000 PC 8000
LDA 2050h
MOV B A
LDA 2052h
MOV C A
MOV A B
SUB C
STA 2054h
LDA 2051h
MOV B A
LDA 2053h
MOV C A
MOV A B
SBB C
STA 2055h
HLT
Starting Address : 8000
Comments :
User Data Grid
Address Data
2050 34
2051 12
2052 11
2053 01
2054 23
2055 11
Hex Code Grid
Address Data
8000 3A
8001 50
8002 20
8003 47
8004 3A
8005 52
8006 20
8007 4F
8008 78
8009 91
800A 32
800B 54
800C 20
800D 3A
800E 51
800F 20
8010 47
8011 3A
1.Data Transfer Group :
MOV MVI LDA STA
LHLD SHLD LXI XCHG
LDAX STAX IN OUT
2.Arithmetic Group :
ADD ADI ADC ACI
SUB SUI SBB SBI
INR DCR INX DCX
DAA
DAD
3.Logical Group :
ANA ANI XRA XRI CMA
ORA ORI CMP CPI CMC
RLC RRC RAL RAR STC
4.Branch and Machine Control Group
JMP J_ CALL C_
RET R_ EI DI PUSH
PCHL RST RIM SIM POP
XTHL SPHL NOP HLT
Auto Insert
Insert
Cancel
Instruction Information :
Code Pad